Boston College, Department of Physics, Fall 2022
PHYS451501: The Physics of Conventional and Alternative Energy
(Updated: 30 August 2022)

Scheduled class meeting times:
Tuesday 1:30-2:45 PM and Thursday 1:30-2:45 PM; Room = Higgins Hall 260

Required Text: Richard A. Dunlap "Sustainable Energy" 2nd Edition. Cengage Learning, Stamford, CT, USA, 2019 (ISBN-13: 978-1337551663; ISBN-10: 133755166X). This book is available in BC bookstore.

Required Text: Bill Gates, "How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need" Alfred Knopfe, Toronto, 2021. This book is readily available on Amazon.

Suggested Reading: (Available in O'Neill Library)
"Sustainable Energy: Choosing Among Options" J. W. Tester, *et al.* 2nd Ed. Massachusetts Institute of Technology Press, 2012. (ISBN = 978-0-262-017477-3).

"Fundamentals of Materials for Energy and Environmental Sustainability" Ed. by D. S Ginley and D. Cahen, Cambridge University Press, 2012, (ISBN = 978-1-107-00023-0).

"Renewable Energy: Power for a Sustainable Future" Ed., Godfrey Boyle, 3rd Edition, Oxford University Press, 2012 (ISBN = 978-0-19-954533-9).

Instructor:
Fr. Cyril P. Opeil, S.J., Ph.D.
Associate Professor of Physics
Office: Higgins 330J, 617-552-8450
Laboratory: Higgins 130, 617-552-3589
Office: St. Mary's Hall, 617-552-8200

Office Hours: As posted at 330J:
Tuesday 3:15-5:00 PM
Thursday 3:15-5:00 PM

The most efficient way to contact Fr. Opeil, S.J., outside of class and office hours, is using email at: <opeil@bc.edu>.

Graduate Assistants/Home Work Grader:
A graduate student *will be* assigned to this class as a teaching assistant. Their primary job is to grade homework assignments, to provide assistance on solving homework problems upon request, and assist the instructor in correcting the quizzes & exams. If you believe that homework is graded unfairly, please see the instructor.

Pre-requisites: A working knowledge of integral, differential calculus and vector algebra will be assumed. This course is designed for upper classmen and foreign exchange students.

Grading:	Homework	12.5 %
	Reading Quizzes	12.5 %
	Mid-term Exam	25 %
	Energy Project Presentation	5% + 5% + 15 %
	Final Exam	25 %

Course Description: PHYS451501 is a three-credit, elective course designed for junior and senior Physics majors. The course consists of an overview of the entire field of energy, production, efficiency and use presented with attention to the mathematical, physical and economic principles needed to assess the feasibility and desirability of each source. This course will provide the most detail with respect to renewable sources (e.g. solar, wind, thermoelectric and geothermal), but will include the topics on nuclear power and fossil fuels. The course assumes an understanding of basic physics, and it strives to address energy problems on a mathematical level and assumes on the part of the student a knowledge of integral and differential calculus.

Course Objectives: This course introduces students to the mathematical, physical, environmental and economic principles of renewable and nonrenewable energy sources. The objective is to give students enough information to allow them to understand the imperative for developing alternative energy resources aside from fossil fuels, to assess the relative merits of various renewable energy sources, perform calculations pertinent to developing such resources, and to understand the factors in optimizing their design, application and locations. The course will stress how to draw conclusions from various statistics and data concerning energy resources.

Course Goals:
1) To understand the critical role different types of energy resources and our use of them affect our modern society and personal lives.
2) To develop a deeper understanding how physics and physics research can affect the development of new and renewable energy resources.
3) Through attending lectures, performing homework problems, reading book research journal articles, working on team projects and preparing presentations, students are expected to gain a greater knowledge of how current energy needs will be met by future energy resources.

Attendance and Course Participation: Continuity and active participation are crucial to the success of this course. Students are expected to participate in class discussions and exercises. Do not hesitate to ask questions in class. You are responsible for all information from each class lecture session whether you attend or not. Absences due to (excused absence) sports activities are to be discussed with the instructor prior to the planned absence. At the **discretion of the instructor**, if a student accumulates more than four unexcused absences from the course lectures this can result in a grade of "F" in this course. Attendance at the lectures will be monitored.

Academic Honesty: Academic honesty is expected at all times in accordance with published Boston College policies. Discussion of the concepts and problem-solving

techniques between students, and with the instructor and homework grader, is strongly encouraged. However, under **no circumstances is it acceptable to copy and turn in solutions taken from ANY other source** – all work must be your own. Please review the Boston College Standards of Academic Integrity that are published in the university catalog and on the web at:
https://www.bc.edu/bc-web/schools/mcas/departments/political-science/undergraduate/academic-integrity.html

It is *your responsibility* to ensure that all work you submit is in accordance with university policies. If you have any questions, please consult with the instructor. Violations will be reported to the Deans' Office and reviewed by the College's Committee on Academic Integrity. Sanctions for violating this policy include the possibility of failing this course with a necessity of repeating this course or a similar level course.

Dropbox: Students need to have their own personal Dropbox account. A Dropbox account is free on the www. Notification of course materials will be sent to your email account and available for downloaded via a Dropbox link. Materials and articles are provided for educational purposes directly to the students at no cost to the student or profit to the instructor and thereby not subject to restrictions of copyright infringement. All journal articles used in this course are available from the BC library journal data base resources.

Reading Assignments: Scientific articles are scheduled to be read throughout this course. A brief multiple-choice question quiz "Reading Quiz" will be given on each reading according to the schedule given below. Copies of assigned articles are available (only to students enrolled in this class, at no cost, for educational purposes to avoid copyright infringement) on Dropbox.

Required Reading: A schedule of required readings are listed in the syllabus that are to be read prior to attending class. These readings appear in the required textbook and are supplemented by articles available on Dropbox.

Homework Exercises: Expect homework assignments each week, note particular schedule below. Homework problems (and eventually solutions) will be available on Dropbox. Student generated homework solutions are due on Monday afternoon in the instructor's office (Higgins 330J) by 5:00 PM. Homework turned in after that time, e.g. 5:01 PM can be graded, at the Instructor's discretion to be worth a maximum of 50% of the normal credit. Homework solutions will be available to the students (normally on the Wednesday) after homework assignments are collected.

In order to receive FULL CREDIT for your problem solutions you must *fully explain* your solutions: briefly outline the logic of your approach to the solution in words, and write out all equations used to obtain your solution. For multiple-questions, the answer requires more than a letter choice, you must include an explanation why that choice is correct. Please be neat: if it cannot be read, it cannot be grade it properly. Homework exercises are worth 10 points each and are given partial credit for incomplete answers. ***Students are strongly encouraged to discuss problems with the instructor*** – or

the graduate assistant during office hours – as posted. The course calendar/schedule below lists when assignments are due. ***Solving the problem set is not the equivalent of copying the problems from another student's solution set.***

For problem set details see schedule below. Corrections and emendations to the homework schedule may be made by the instructor as the course progresses.

Midterm Exam: The midterm exam will consist of an 8-10 page essay/paper due on 01 November 2022. Send the essay in pdf format to opeil@bc.edu.

The essay should answer the question: *What argument can be made for developing alternative energy resources given the considerations of future climate change?*

A primary resource for this essay will be:
Bill Gates, "How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need" Alfred Knopfe, Toronto, 2021. (Chapter 2, 3, 4 and 7).

Use research articles available in the library data bases and the Course Reserves for PHYS451501 in O'Neill Library. Endnotes are preferred for citing research works.

Final Exam: One written final exam (two hours) will be given according to the published MCAS schedule. Final exams cover **all** material discussed in class lectures, example problems, assigned readings in the textbook and problems assigned for homework. However, the exams **do not** include the material from the Reading Assignments that are evaluated only by weekly quizzes.

Every student must take the final exam at the officially scheduled time, exceptions are granted rarely for exceptional reasons and must be discussed with the instructor one week prior to the scheduled exam.

Energy Project Presentations:

As a way of actively engaging questions and gaining experience about energy and renewable resources, students will give PowerPoint presentations during the last week(s) of class. PowerPoint presentations will be given by students according to the syllabus schedule. Each presentation will be 12-15 minutes long with 2-3 minutes for questions. A list of possible presentation topics will be available via Dropbox. Email opeil@bc.edu with your topic choice. First come, first serve. If you want to suggest a project not listed here, contact the instructor to discuss the project. The deadlines for choosing a topic and for sending the completed PowerPoint presentation to the instructor are given in the syllabus schedule. Students are required to discuss the project with the instructor. Such a discussion normally occurs during office hours of the instructor. Such discussions are to review the resources available for a particular project.

The presentation will be given orally in class. The final project submission will be a video and audio PowerPoint or Keynote file so that the presentation can be viewed by anyone after the conclusion of the course.

Energy Project Presentations will be judged by the instructor and are assigned a grade. The score from the Energy Project will be 25% of your final course grade.

+++ 0-5 points are awarded upon discussion and approval of your project with the instructor.
+++ 0-5 points are awarded for handing-in the completed project by the deadline.
+++ 0-15 points are awarded for the quality of the presentation given.

The criteria upon which your presentation will be judged are:
a) Description of the project undertaken.
b) Clarity of issue presented.
c) Care in presentation of the data or outcome.
d) What was the project and what was learned from the project?
e) How does this issue affect your knowledge about energy and renewable energy?
f) Use of research by other scientists for comparison?

Disability Accommodation: If you have a disability and will be requesting a disability accommodation for this course, please inform the instructor. Information concerning disability accommodation can be found by contacting Paulette Durrett the Assistant Dean for Students with Disabilities (durrett@bc.edu) 617-552-3470. Further information can be found on the www link:

http://www.bc.edu/sites/accessibility/accommodations/for-students/disability-services.html

Eating and Drinking: *Absolutely* no food or beverages are to be brought into or consumed in the lecture room. Not consuming of food and or beverages in the lecture hall is a matter of simple courtesy and civility toward the instructor and your fellow students.

Cell Phone Use: Students are required to turn off their cell phones prior to attending class. It is a matter of courtesy to the instructor and other students that cell phones do not ring during class time.

Computer Use During Lectures: Students are asked not to use personal computers during the lecture times. Listen, take notes and ask questions.

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Semester Schedule for Fall 2022

T = Tuesday, class time 1:30-2:45 PM
TH = Thursday, class time 1:30-2:45 PM
M = Monday, Homework due 5:00 PM, Higgins 330J
Key: L# = Lecture, RA# = Reading Assignment, HW# = Homework
Sustainable Energy, by Richard A. Dunlap, 2nd Edition = Dunlap Section #

2022 August
29 Monday - Fall 2022 classes begin
30 Tuesday - Lecture01, Syllabus and Course Introduction

2022 September

01 Thursday - Lecture02, The History of Energy Use
no Reading Quiz scheduled
Required reading 2nd ed., Dunlap, Sections 1.1-1.3 & 1.7

05 Monday - no classes - Labor Day

06 Tuesday - Lecture03, Petroleum I
Required reading 2nd ed., Dunlap, Sections 2.1-2.6

08 Thursday - Lecture04, Petroleum II
Reading Quiz01
Required reading 2nd ed., Dunlap, Sections 3.1-3.6a & 3.7
Mass of the Holy Spirit, noon-1:15 PM, lecture time: 1:30-2:45 PM

12 Monday - HW01 due in Higgins 330 J and
Library Periodical Search Exercise due

13 Tuesday - Lecture05, Engines, Generators and Thermodynamics I,
Required Reading: D. E. Cole "The Wankel Engine"
Scientific American **227**, 14 (August 1972).
Required reading 2nd ed., Dunlap, Sections 1.4-1.5

15 Thursday - Lecture06, Engines, Generators and Thermodynamics II,
Required Reading: Dunlap Section 17.4
Reading Quiz02

19 Monday - HW02 due

20 Tuesday - Lecture07, Gasoline Alternatives/Electric Cars I,
Required Reading: Dunlap Sections 17.8-17.9

22 Thursday - Lecture08, Gasoline Alternatives/Electric Cars II
Required Reading: Dunlap Sections 20.2-20.5
Reading Quiz03,

26 Monday - HW03 due

27 Tuesday - Lecture09, Coal, Tar Sands, Natural Gas I
Required Reading: N. de Nevers "Tar Sands and Oil Shales"
Scientific American 214, 21 (Feb. 1966).

29 Thursday - Lecture10, Coal, Tar Sands, Natural Gas II
Required Reading: Dunlap Sections 3.8-3.10
Reading Quiz04

2022 October

03 Monday - HW04 due

04 Tuesday - Lecture11, Nuclear Energy I
Required Reading: J. A. Lake, et al., "Next-Generation Nuclear Power"
Scientific American 286, 72-81 (Jan. 2002).
Required Reading: 2nd edition Dunlap, Sections 5.1-5.5.

06 Thursday - Lecture12, Nuclear Energy II
Required Reading: 2nd edition Dunlap, Sections 6.1-6.4, 7.1-7.3.
Reading Quiz05

10 Monday Fall Break—No classes

11 Tuesday (Substitute Monday class schedule) no class for PHYS451501 scheduled.

13 Thursday - Lecture13, Thermoelectrics I
Required Reading: D. M. Rowe, "Thermoelectrics, an Enviromentally-Friendly

Source of Electrical Power" *Renewable Energy* **16**, 1251 (1999).
Reading Quiz06

17 Monday - HW05 due

18 Tuesday - Lecture14, Thermoelectrics II

20 Thursday - Lecture15, Wind Energy I
Required Reading: Dunlap Sections 10.1-10.5
Reading Quiz07
Deadline for choosing Energy Project Topic at 5:00 PM

24 Monday - HW06 due

25 Tuesday - Lecture16, Wind Energy II
Required Reading: Kumar et al., A critical review of vertical axis wind turbines for urban applications, Renewable and Sustainable Energy Reviews 89 (2018) 281-291.

27 Thursday - Lecture17, Solar Energy I
Required Reading: Dunlap Sections 9.1-9.6
Reading Quiz08

31 Monday - HW07 due

2022 November

01 Tuesday - Lecture18, Solar Energy II
Midterm Exam Due

03 Thursday - Lecture19, Geothermal Energy
Required Reading: Dunlap Sections 15.1-15.6
Reading Quiz09

07 Monday - HW08 due

08 Tuesday - Lecture20, Material Recycling and Trash to Cash
Required Reading: Dunlap Sections 16.6-16.7

10 Thursday - Lecture21, Biomass Fuel for Residential Heating
Required Reading: Dunlap Sections 16.1-16.5
Reading Quiz10

14 Monday - HW09 due

15 Tuesday - Lecture22, Insulation and Cooling I
Required Reading: Dunlap Sections 17.1-17.7

17 Thursday - Lecture23, Harvesting Ocean Wave Energy
Required Reading: Dunlap Sections 14.1-14.8
Optional Reading: J. F. Cabeza, *et al., Energy and Buildings* **42**, 630 (2010).
Reading Quiz11

21 Monday - HW10 due

22 Tuesday - Lecture24, Third World Energy Devices

24 Thursday - Thanksgiving Holiday, no class for PHYS 451501 scheduled.

28 Monday - HW11 due

29 Tuesday - Lecture25, Climate Change Phenomenon
Energy Presentation Slides & Audio to Instructor by 5:00 PM

2022 December

01 Thursday - Lecture26, Care for Our Common Home
Required Reading: Laudato Si' Pope Francis (2015) para. 1, 13-15, 19-26, 27-31, 32-42, 93-95, 102-105, 156-158.

Reading Quiz12
06 Tuesday - Lecture27 - Student Energy Presentations (3-4, 20 min. presentations)
08 Thursday - Lecture28 - Student Energy Presentations (3-4, 20 min. presentations)
13 Tuesday - Study days
14 Wednesday Term examinations through 12/21, no Sunday exams
Check the MCAS final exam schedule for PHYS451501.

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2022 Reading Quiz Assignments – PHYS451501

RA01-Energy Crossroads & Peak Oil (www youtube videos):
(2 min, 35 sec) https://www.youtube.com/watch?v=OHBM9OS0cOU
(5 min, 30 sec) https://www.youtube.com/watch?v=Fjn8hja6HRg
(5 min, 32 sec) https://www.youtube.com/watch?v=suREHgvz5yE

RA02-H. L. MacLean and L. B. Lave, "Evaluating automobile fuel/propulsion system technologies" P*rogress in Energy and Combustion Science* **29**, 24-33 (only section 4), (2003).

RA03-D. G. Wilson, "Alternative Automobile Engines" *Scientific American* **239**, 39 (1978).

RA04-J. Singh, and S. Gu, "Commercialization potential of microalgae for biofuels production" *Renewable and Sustainable Energy Reviews*, **14**, 2596 (2010).

RA05-A. Raukas and J. - M. Punning, "Environmental problems in the Estonian oil shale industry" *Energy & Environ. Sci.,* **2**, 723-728 (2009); The presentation: *Estonian Solid Heat Carrier Technology (TSK) as a Development Platform for Oil Shale Processing* from EESTI ENERGIA.

RA06- D. Mallants, J. Marivoet and X. Sillen, "Performance assessments of the disposal of vitrified high-level waste in a clay layer" *Journal of Nuclear Materials*, **298**, 125 (2001).

RA07- S. B. Riffat, X. Ma "Thermoelectrics: a review of present and potential applications" *Applied Thermal Engineering* **23**, 913 (2003).

RA08-J. D. Booker, et al., "A compact, high efficiency contra-rotating generator suitable for wind turbines in the urban environment" *Renewable Energy* **35**, 2027 (2010).

RA09-D. Mills, "Advances in solar thermal electricity technology" *Solar Energy* **76,** 19 (2004).

RA10- G. W. Huttrer, "Geothermal Heat Pumps: An Increasingly Successful Technology" R*enewable Energy* **10**, 481 (1997).

RA11-C. J. Rydh and M. Karlstrom "Life cycle inventory of recycling portable nickel–cadmium batteries" *Resources, Conservation and Recycling* **34**, 289 (2002).

RA12-T.M.I. Mahlia et al. "Correlation between thermal conductivity and the thickness of selected insulation materials for building wall" *Energy and Buildings* **39**, 182 (2007).

RA13- J.-M. Tarascon, M. Gratzel, "Research Initiatives Towards Sustainable Handling of Energy" in Materials for Sustainable Energy (Macmillan Publishers Ltd.: London, 2011).

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Course Reserves Available for PHYS451501 in O'Neill Library:

Richard A. Dunlap "Sustainable Energy" 2nd Edition. Cengage Learning, Stamford, CT, USA, 2019 (ISBN-13: 978-1337551663; ISBN-10: 133755166X).

J. W. Tester, *et al.* "Sustainable Energy: Choosing Among Options" 2nd Ed. Massachusetts Institute of Technology Press, 2012. (ISBN = 978-0-262-017477-3)

D. S Ginley and D. Cahen, "Fundamentals of Materials for Energy and Environmental Sustainability" Cambridge University Press, 2012. (ISBN = 978-1-107-00023-0)

Godfrey Boyle "Renewable Energy: Power for a Sustainable Future" 3rd Edition, Oxford University Press, 2012. (ISBN = 978-0-19-954533-9)

Walter Witzel and Dieter Seifried, "Renewable Energy - The Facts" Taylor and Francis Publishing, 2010. (ISBN: 978-1-84971-159-3)

F. A. Farret and M. Godoy Simoes "Integration of Alternative Sources of Energy" Wiley-Interscience Publication, John Wiley and Sons Inc., IEEE Press, 2006. (ISBN-13:978-0-471-71232-9)

Carla S. Jones and Stephen P. Mayfield, "Our Energy Future: Introduction to Renewable Energy and Biofuels" University of California Press, 2016. (ISBN: 978-0-520-27877)

Robert A. Ristinen and Jack J. Kraushaar, 2nd ed., Energy and the Environment John Wiley and Sons, New York, 2006.

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PHYS451501: Physics of Conventional and Alternative Energy
Homework Assignments - Fall 2022

Directions:

This page is the first page of every homework assignment you hand in for credit. Put your name in the space below and circle the correct homework assignment number below. Each problem normally is worth 10 points for a complete solution. Homework assignments are to be delivered to Higgins 330J by 5 PM on the designated days. The list of homework assignments from the textbook and other creative problems are available on Dropbox.

Name: _____

Homework Assignment: (circle one below)

01 02 03 04 05 06 07 08 09 10 11

M-12Sept	HW01 due
M-19Sept	HW02 due
M-26Sept	HW03 due
M-03Oct	HW04 due
M-17Oct	HW05 due
M-24Oct	HW06 due
M-31Oct	HW07 due
M-07Nov	HW08 due
M-14Nov	HW09 due
M-21Nov	HW10 due
M-28Nov	HW11 due

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